A Lightweight CAN Bus IDS Based on ACK-Bit Differential Voltage*

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Abstract

This paper proposes a lookup-table-based physical-layer intrusion detection system for detecting unauthorized CAN node insertion using ACK-bit differential voltage. The proposed method estimates the number of connected nodes from the measured voltage and detects insertion when the estimated node count exceeds the predefined normal count. A controlled CAN bus testbed was configured with 3 to 11 nodes, and 50 voltage measurements were collected for each node-count condition. The median voltages and the midpoints between adjacent medians were used to construct the lookup table. Two experiments with 100 estimation trials per evaluated condition achieved 100% node-count estimation accuracy under controlled testbed conditions.

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